

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	T. Varun Tej	Reg. No.:	192511217
Section / Batch:		Date:	31/08/2026

Code of Conduct

I T. Varun Tej/(192511217) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Varun Tej

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks**Q2** ✓
Graph - Topological Sort
20 marks**Q3** ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks**Q4** ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks**Q5** ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- a. All zones must be connected
- b. Total installation cost must be minimized
- c. No redundant connections (no cycles) allowed
- d. The network must remain fully connected

Your AnswerKruskalMST(G)

Input:

$G = (V, E)$

V = set of zones (vertices)

E = set of cable connections (edges)

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks**Q2** ✓
Graph - Topological Sort
20 marks**Q3** ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks**Q4** ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks**Q5** ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q2 Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- a. Detect cycle before sorting
- b. Use DFS approach

Your Answer

TopologicalSort(G)

Input:

$G = (V, E)$, a directed graph

Output:

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QUESTIONS
Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- a. Use MST for infrastructure
- b. Use shortest path for routing

Your Answer

MST_Dijkstra(G, Source, Destination)

Input:

G = (V, E), transportation network

V = set of locations

E = roads with weights (cost/distance)

 **Save Answer**

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QUESTIONS
Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q4 Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Shortest Path with "Toll-Free" Voucher; You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

DijkstraWithVoucher(G, S, D)

Input:

G = (V, E) weighted graph

S = Source vertex

D = Destination vertex

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks**Q2** ✓
Graph - Topological Sort
20 marks**Q3** ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks**Q4** ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks**Q5** ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

5/5 answered

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```
SecondBestMST(Graph G):  
    # Step 1: Find MST  
    MST = Kruskal(G) # or Prim(G)  
    W = weight(MST)  
    best_second = ∞
```

[Save Answer](#)